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PRACTICAL WEEK 3

Task 1: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Source code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    printf("Parent process started.\n");
    printf("Parent PID  : %d\n", getpid());
    printf("Parent PPID : %d\n\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("----- CHILD PROCESS -----\n");
        printf("Child PID  : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("Child process is running...\n");

        sleep(5);

        printf("Child process completed.\n");
        exit(0);
    }
    else
    {
        // Parent process
        printf("----- PARENT PROCESS -----\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID  : %d\n", pid);

        printf("Parent is waiting for child...\n");

        wait(NULL);

        printf("Child process has terminated.\n");
        printf("Parent process completed.\n");
    }

    return 0;
}
```

1) To save press control O
2) To come back to shell press control X
3) Then DO compilation

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano practical3.1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc practical3.1.c -o practical3.1
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./practical3.1
Parent process started.
Parent PID  : 550
Parent PPID : 342

----- PARENT PROCESS -----
Parent PID : 550
Child PID  : 551
Parent is waiting for child...
----- CHILD PROCESS -----
Child PID  : 551
Child PPID : 550
Child process is running...
Child process completed.
Child process has terminated.
Parent process completed.
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process started.\n");
    printf("PID = %d\n", getpid());

    printf("Process is running...\n");

    for (long i = 0; i < 1000000000L; i++)
    {
        // Busy loop
    }

    printf("Process is going to sleep...\n");

    sleep(10);

    printf("Process completed.\n");

    return 0;
}
```

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano practical3.2.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc practical3.2.c -o practical3.2
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

OUTPUT:

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano practical3.2.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc practical3.2.c -o practical3.2
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./practical3.2
Process started.
PID = 603
Process is running...
Process is going to sleep...
Process completed.
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```